

Subject Description Form

Subject Code	EE570
Subject Title	Design and Analysis of Smart Grids
Credit Value	3
Level	5
Pre-requisite / Co-requisite/ Exclusion	Nil
Objectives	1. To provide students with a comprehensive understanding on design and analysis of smart grids; 2. To ensure the students are aware of the current state-of-the-art on design, operation and control of smart grid; 3. To acquire knowledge on the components in smart grids and their functions; and 4. To enable students to apply advanced analysis tools in planning and operation of smart grids.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Acquire in-depth understanding on the recent development of power grids, i.e. smart grid; b. Apply advanced analysis tools in planning and operation of smart grids; and c. Acquire skills in presentation and interpretation of results in written form.
Subject Synopsis/ Indicative Syllabus	1. Introduction to smart grid: Overview of power system operation; Comparison between existing grid and smart grid; Objectives; Benefits; Challenges; Basic structure and functions of components. 2. Communications and measurement: Latest technologies; Wide Area Monitoring Systems (WAMS), Phasor Measurement Units (PMU), Smart Meters, Smart Appliances, and Advanced Metering Infrastructure (AMI); GIS and Google Mapping Tools; Multiagent Systems Technology. 3. Micro-grid: Concept of micro-grid; design and analysis; distributed generation; distributed automation. 4. Renewable energy and storage: Renewable energy resources and options for smart grid including solar energy, wind energy, fuel cell, biomass etc.; Penetration and variability; Demand Response; Electric vehicles and plug-in hybrid; Battery energy storage systems. 5. Interoperability, standards and cyber security: State-of-the-art, Benefits, Challenges, Risks. 6. Analysis tools: Power/load flow studies; Static security assessment; State estimation and stability assessment; Reliability assessment; Decision support tools; Advanced optimization and control; Environmental impacts; Pathway for designing smart grid. 7. Standards and critical infrastructure protection: State-of-the-art, Benefits, Challenges, Risks.

Teaching/Learning Methodology	Lectures and tutorials are the primary means of conveying the concepts and theories. Mini-projects are designed to supplement the lecturing materials so that the students are encouraged to take extra readings and the latest development of the smart grids.				
	Teaching/Learning Methodology	Outcome			
		a	b	c	
	Lectures	✓	✓		
	Tutorials		✓	✓	
	Mini-project		✓	✓	
Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed		
			a	b	c
	1. Examination	63%	✓	✓	
	2. Class test	18%	✓	✓	
	3. Mini-project	19%		✓	✓
	Total	100%			
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:				
	The outcomes of understanding the development of the smart grid and the application of advanced analysis tools are assessed by the usual means of examination and tests. Mini-projects and written reports assess those analytical skills, problem-solving techniques and technical reporting.				
	Student Study Effort Expected	Class contact:			
		▪ Lectures			36 Hrs.
▪ Tutorial			3 Hrs.		
Other student study effort:					
▪ Self-study			50 Hrs.		
▪ Mini-project			16 Hrs.		
Total student study effort			105 Hrs.		
Reading List and References	1. P. Sioshansi, “Smart Grid: Integrating Renewable, Distributed & Efficient Energy,” Elsevier Inc., 2012. 2. J.A. Momoh, “Smart Grid: Fundamentals of Design and Analysis,” 2012 IEEE, John Wiley & Sons, Inc., 2012. 3. Peter Fox-Penner, “Smart Power: Climate Change, the Smart Grid, and the Future of Electric Utilities,” Island Press, 2010.				